

Deciding When to Submit a 510(k) for a Change to an Existing Device

Guidance for Industry and Food and Drug Administration Staff

This guidance represents the current thinking of the Food and Drug Administration (FDA or Agency) on this topic. It does not establish any rights for any person and is not binding on FDA or the public. You can use an alternative approach if it satisfies the requirements of the applicable statutes and regulations. To discuss an alternative approach, contact the FDA staff or Office responsible for this guidance as listed on the title page.

I. Introduction

Almost from the enactment of the Medical Device Amendments to the Federal Food, Drug, and Cosmetic Act (the FD&C Act) in 1976, the Food and Drug Administration (FDA or the Agency) has attempted to define with greater clarity when a change in a medical device would trigger the requirement that a [manufacturer](#) submit a new premarket notification (510(k)) to the Agency. This document supersedes FDA's guidance *Deciding When to Submit a 510(k) for a Change to an Existing Device (K97-1)*, issued on January 10, 1997. This guidance is not intended to implement significant policy changes to FDA's current thinking on when submission of a new 510(k) is required. Rather, the intent of this guidance is to enhance the predictability, consistency, and transparency of the "when to submit" decision-making process by providing a least burdensome approach, and describing in greater detail the regulatory framework, policies, and practices underlying such a decision.

For the current edition of the FDA-recognized standards referenced in this document, see the FDA Recognized Consensus Standards Database at <http://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfStandards/search.cfm>.

FDA's guidance documents, including this guidance, do not establish legally enforceable responsibilities. Instead, guidances describe the Agency's current thinking on a topic and should be viewed only as recommendations, unless specific regulatory or statutory requirements are cited. The use of the word *should* in Agency guidance means that something is suggested or recommended, but not required.

II. Background

The regulatory criteria in 21 CFR 807.81(a)(3) state that a premarket notification must be submitted when:

(3) The device is one that the person currently has in commercial distribution or is reintroducing into commercial distribution, but that is about to be significantly changed or modified in design, components, method of manufacture, or intended use. The following constitute significant changes or modifications that require a premarket notification:

(i) A change or modification in the device that could significantly affect the safety or effectiveness of the device, e.g., a significant change or modification in design, material, chemical composition, energy source, or manufacturing process.

(ii) A major change or modification in the intended use of the device.

FDA issued the original guidance *Deciding When to Submit a 510(k) for a Change to an Existing Device (K97-1)* on January 10, 1997 to provide guidance on this regulatory language. As stated in that guidance, the key issue in the interpretation of 21 CFR 807.81(a)(3) is that the phrase “could significantly affect the safety or effectiveness of the device” and the use of the adjectives “major” and “significant” sometimes lead FDA and device manufacturers to different interpretations. The original guidance provided the Agency’s interpretation of these terms, with principles and points for manufacturers to consider in analyzing how changes in devices may affect safety or effectiveness and determining whether a new 510(k) must be submitted for a particular type of change. The current guidance preserves the basic format and content of the original, with updates to add clarity. The added clarity is intended to increase consistent interpretations of the guidance by FDA staff and manufacturers and provide a more transparent framework for determining when submission of a new 510(k) is required.

The 510(k) Process and the Quality System Regulation

Any guidance on 510(k)s for changes to a legally marketed device should consider the role the Quality System (QS) regulation, 21 CFR Part 820, plays in changes to devices. For some types of changes to a device, the Agency believes that submission of a new 510(k) is not required and that reliance on existing QS requirements is the least burdensome approach to reasonably assure the safety and effectiveness of the changed device.

Regardless of whether a change requires premarket review, the QS regulation requires manufacturers of finished medical devices to review and approve changes to device design and production (21 CFR 820.30 and 820.70) and document changes and approvals in the device master record (21 CFR 820.181). Any process whose results cannot be fully verified by subsequent inspection and testing must be validated (21 CFR 820.75), and changes to the process require review, evaluation, and revalidation of the process where appropriate (21 CFR 820.75(c)).

The net effect of the QS regulation is to require that, when manufacturers of a finished medical device make a change in the design of a device, there is a process in place to demonstrate that the

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manufactured device meets the change in design specifications (or the original specifications, if no change was intended). They must keep records, and these records must be made available to an FDA investigator upon request (see Section 704(e) of the FD&C Act). For many changes to a device, submission of a new 510(k) may not be required. In these cases, including for many design changes, compliance with the QS regulation can reasonably assure the safety and effectiveness of the changed device.

Least Burdensome Principles

The least burdensome provision concerning 510(k)s states that FDA “shall only request information that is necessary...” and “shall consider the least burdensome means of demonstrating substantial equivalence...” (see section 513(i)(1)(D)(i) of the FD&C Act). While not changing the standard for substantial equivalence, this provision states that FDA shall only request the “minimum required information” necessary to support a determination of substantial equivalence (see sections 513(i)(1)(D)(ii)-(iii) of the FD&C Act). The recommendations discussed in this guidance for evaluating when a change in a medical device would trigger the requirement that a manufacturer submit a new 510(k) to the Agency are consistent with least burdensome principles, and applies them in discussing the considerations that may affect the decision-making about when to submit a new 510(k) for a device change or modification.

III. Scope

This guidance will aid manufacturers of medical devices subject to premarket notification requirements who intend to modify a 510(k)-cleared device (or group of devices) or other [device](#) subject to 510(k) requirements, such as a preamendments device or a device that was granted marketing authorization via the De Novo classification process¹ under section 513(f)(2) of the FD&C Act (also referred to together as “existing devices”), during the process of deciding whether the change exceeds the regulatory threshold of 21 CFR 807.81(a)(3) for submission and clearance of a new 510(k). Note that any person required to register under 21 CFR 807.20 who plans to introduce a device into commercial distribution for the first time must, per 21 CFR 807.81(a)(2), submit a 510(k) if that device is not exempt from premarket notification requirements. Also note that devices with changes requiring submission of a new 510(k) may not be legally commercially distributed before FDA clears the changed device (21 CFR 807.100(a) and sections 513(f)(1) and 513(i) of the FD&C Act). This guidance is not intended to address changes to devices that are 510(k)-exempt or that require premarket approval (PMA). Also, the scope of key terminology used in this guidance, particularly intended use and indications for use, is limited to medical devices and not other FDA-regulated products.

This document incorporates concepts and recommendations from existing FDA guidance and policy, such as *Submission and Review of Sterility Information in Premarket Notification (510(k)) Submissions for Devices Labeled as Sterile* (<http://www.fda.gov/downloads/MedicalDevices/.../ucm109897.pdf>), and device-specific final guidance documents that identify and characterize specific scenarios regarding when submission of new 510(k)s are required or not required based on changes to an existing device. In some

¹ This guidance applies to devices granted marketing authorization via the De Novo classification process that are not exempt from premarket notification requirements.

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cases, FDA's thinking has derived from its experience in situations involving only a few manufacturers of a limited number of devices. In such instances, we have attempted to generalize the concepts to apply to a broader range of devices. However, special cases exist where FDA has established definitive final guidance for changes to specific devices, e.g., FDA's guidance on daily wear contact lenses, *Premarket Notification (510(k)) Guidance Document for Daily Wear Contact Lenses*

(<http://www.fda.gov/MedicalDevices/DeviceRegulationandGuidance/GuidanceDocuments/ucm080928.htm>). This guidance is not intended to supersede such final device-specific guidance but may cover areas not addressed in such device-specific guidance.

Recalls: This guidance is also intended to apply to situations when a legally marketed existing device is the subject of a recall, correction, or removal, and a change in the device or its [labeling](#) is necessary. For more information on recommended procedures in a recall situation, please see *Blue Book Memorandum K95-1, 510(k) Requirements During Firm-Initiated Recalls* (<https://www.fda.gov/MedicalDevices/DeviceRegulationandGuidance/GuidanceDocuments/ucm080297.htm>). As stated in that guidance, if a correction alters a device rather than simply restoring it to its original specifications, submission of a new 510(k) may be required. FDA may use this guidance in determining whether submission of a new 510(k) is warranted in cases where the correction does alter the device.

Private Label Distributors and Repackagers: Private label distributors and repackagers are exempt from submitting a 510(k) if they satisfy the requirements of 21 CFR 807.85(b).

Software Changes: This guidance does not address [software](#) changes or modifications. Please refer to FDA's guidance *Deciding When to Submit a 510(k) for a Software Change to an Existing Device*

(<https://www.fda.gov/downloads/MedicalDevices/DeviceRegulationandGuidance/GuidanceDocuments/UCM514737.pdf>) for recommendations regarding software changes.

This guidance does apply to non-software changes to devices containing software and non-software changes to software that is a medical device on its own. For example, labeling changes to software are covered by Section A of this guidance, and non-software technology changes and materials changes to existing devices that contain software are covered by Sections B through D of this guidance.

When there are multiple changes that affect labeling or hardware in addition to software, the manufacturer should assess the changes using both the general and software-specific modifications guidances. If use of either guidance leads to a "New 510(k)" conclusion, submission of a new 510(k) is likely required.

Combination Products: This guidance does not specifically address combination products, such as drug/device or biologic/device combinations; however, the general principles and concepts described herein may be helpful to manufacturers in determining whether submission of a 510(k) is required for changes to device constituent parts of combination products.

Remanufactured or Reprocessed Single Use Devices: This guidance is not intended to address whether submission of 510(k)s are required from remanufacturers of existing devices who do not [hold the 510\(k\)](#) for the device, such as reproducers of [single-use devices](#). This guidance does apply to reproducers and remanufacturers who hold their own 510(k) and are addressing

changes or modifications. Remanufacturer is defined at 21 CFR 820.3(w) as “any person who processes, conditions, renovates, repackages, restores, or does any other act to a finished device that significantly changes the finished device’s performance or safety specifications, or [intended use](#).”

IV. Guiding Principles

In using this guidance for deciding whether to submit a new 510(k) for a change to an existing device, a number of guiding principles should be followed. Some derive from existing FDA 510(k) policy and are widely known, and others are necessary for using the logic scheme contained in this guidance. Thus, anyone using this guidance should bear in mind the following Guiding Principles:

- 1. Changes made with intent to significantly affect safety or effectiveness of a device –** If a manufacturer modifies their device with the intent to significantly affect the safety or effectiveness of the device (for example, to significantly improve clinical outcomes, to mitigate a known [risk](#), in response to adverse events, etc.), submission of a new 510(k) is likely required. A change *intended* to significantly affect the safety or effectiveness of the device is considered to be a change that “*could* significantly affect the safety or effectiveness of the device” and thus requires submission of a new 510(k) regardless of the considerations outlined below. Changes that are not intended to significantly affect the safety or effectiveness of a device, however, should still be evaluated to determine whether the change could significantly affect device safety or effectiveness.

If a manufacturer modifies their device to address a violation or recall, they should refer to FDA guidances *Blue Book Memorandum K95-1, 510(k) Requirements During Firm-Initiated Recalls*

(<http://www.fda.gov/MedicalDevices/DeviceRegulationandGuidance/GuidanceDocuments/ucm080297.htm>) and *Distinguishing Medical Device Recalls from Medical Device Enhancements* (<http://www.fda.gov/ucm/groups/fdagov-public/@fdagov-meddev-gen/documents/document/ucm418469.pdf>).

- 2. Initial risk-based assessment –** To determine whether a change or modification could significantly affect the safety or effectiveness of a device, the manufacturer should first conduct a risk-based assessment, using the guidance below, of whether the change could significantly affect the device’s safety or effectiveness, either positively or negatively. This risk-based assessment should identify and analyze all new risks and changes in existing risks resulting from the device change, and lead to an initial decision whether or not submission of a new 510(k) is required.

For the purposes of this guidance, we have chosen the term “risk-based assessment” to describe the analysis that should be completed to assist in the determination of whether or not a change could significantly affect safety or effectiveness of the device. Although common risk analysis methods define risk in terms of device harms and their effects on safety, it is important to note that whether submission of a new 510(k) is required depends on whether the change could significantly affect the safety *or effectiveness* of the device. Therefore, manufacturers should also consider the possible effects a device

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change may have on device effectiveness. As such, we have chosen to use the distinct terminology of “risk-based assessment.”

3. **Unintended consequences of changes** – After a manufacturer considers whether the change was made with the intent to significantly affect safety or effectiveness, the manufacturer should also consider whether the change could have unintended consequences. In order to fully assess device changes, manufacturers should consider the effects of the planned device changes and whether these changes create any intended and/or unintended consequences. For example, changes in sterilization may unintentionally affect device materials, or changes to materials may unintentionally affect the performance of the device. Any unintended consequences such as these should be evaluated according to the relevant flowcharts (and their companion text) to determine whether submission of a new 510(k) is required. For instance, a change in sterilization that may unintentionally affect device performance should be reviewed as a sterilization change under B3 and as a performance specification change under B5.
4. **Use of risk management** – A risk-based assessment as referred to throughout this document is based on the combination of multiple risk concepts that are important for managing the risks of medical devices. [Hazards](#) and hazardous situations, risk estimation, risk acceptability, risk control, risk/benefit analysis and overall risk evaluation are all concepts that can be applied during the design and development of a medical device. The concept of risk, as defined in ISO 14971: *Medical devices – Application of risk management to medical devices*, is the combination of the probability of occurrence of [harm](#) and the severity of that harm. Although the risk terminology used in this document is primarily derived from ISO 14971, we recognize that an individual manufacturer’s terminology may differ. Because 21 CFR 807.81(a)(3)(i) requires submission of a new 510(k) when a change “could significantly affect safety or effectiveness,” both safety and effectiveness should be considered in evaluating a device’s risk profile and performing a risk-based assessment, as explained in Section E.

This guidance states throughout that submission of a new 510(k) is likely required when a risk-based assessment of the changed device identifies any new risks or significantly modified existing risks. For the purposes of this guidance, a new risk is a new hazard or hazardous situation that did not exist for the original device (see discussion of appropriate comparative device below) and the pre-mitigation risk level associated with the new risk is not considered to be acceptable. For the purposes of this guidance, a device change could be considered to significantly modify an existing risk if it changes the risk score, risk acceptability category, or duration of risk. See Section E for further explanation.

5. **The role of testing (i.e., verification and validation activities) in evaluating whether a change could significantly affect safety and effectiveness** - If the initial decision following the risk-based assessment is that submission of a new 510(k) is not required, this decision should be confirmed by successful, routine verification and validation activities. If routine verification and validation activities produce any unexpected results, any prior decision that submission of a new 510(k) is not required should be reconsidered, as discussed in **B5.4** for non-IVD devices and **D4** for IVD devices. “Routine” activities in this context refer to the original design verification and validation activities that were done to assess the original device design. Because 21 CFR

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807.81(a)(3) requires submission of a new 510(k) for a change that “*could* significantly affect safety or effectiveness,” if the result of a risk-based assessment is that a change could significantly affect safety or effectiveness, submission of a new 510(k) is required even if routine verification and validation activities are conducted successfully without any unexpected results. Note that verification and validation requirements apply for all devices subject to 21 CFR 820.30, and must be conducted regardless of whether submission of a new 510(k) is required.

- 6. Evaluating simultaneous changes to determine whether submission of a new 510(k) is required** – Because many simultaneous changes may be considered at once, each change should be assessed separately, as well as in aggregate.
- 7. Appropriate comparative device and cumulative effect of changes** – In using this guidance to help determine whether a particular change requires submission of a new 510(k), manufacturers should conduct a risk-based assessment that compares the changed device to their device as previously found to be substantially equivalent in their most recently cleared 510(k), to their preamendments device (if the device was in commercial distribution before May 28, 1976 and there have not been changes to it subsequently cleared in a 510(k)), or to their device that was granted marketing authorization via the De Novo classification process (if there have not been changes to it subsequently cleared in a 510(k)). The appropriate comparative device is referred to as the “original device” throughout this guidance document. Of note, this comparison is different from a substantial equivalence comparison between the modified device and a legally marketed predicate device. Manufacturers may make a number of changes without having to submit a new 510(k), but each time they make a change, the modified device should be compared to the original device (i.e., the device described in their most recently cleared 510(k) for the device, their legally marketed preamendments device, or their device that was granted marketing authorization via the De Novo classification process). When the cumulative effect of individual changes triggers the regulatory threshold for submission, the manufacturer should submit a new 510(k). When it does not, the manufacturer must document the change(s) (see 21 CFR Part 820.30).
- 8. Documentation requirement** – Whenever manufacturers change their device, they must take certain actions to comply with the QS regulation, 21 CFR Part 820, unless the device in question is exempt by regulation from the QS regulation. The QS regulation requires, among other things, that device changes be documented. The scope and type of documentation may vary, but the process of documenting the decisions described in this guidance should be established as part of the manufacturer’s own quality system. See Appendix B for further explanation and recommendations on [documentation](#).
- 9. 510(k) submissions for modified devices** – When a new 510(k) is submitted for a device with multiple changes, that 510(k) should describe all changes that trigger the requirement for submission of a new 510(k). To help ensure that FDA has a complete understanding of the device under review, that 510(k) should also describe other changes since the most recently cleared 510(k) (i.e., those that did not require submission of a new 510(k)) that would have been documented as part of the first 510(k) for that device. For instance, 510(k)s typically include a listing of device [warnings](#) in the labeling, so if a warning in the device’s labeling had been changed, that change should be described in

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the new 510(k), even if that change did not itself trigger the requirement for submission of a new 510(k). However, a 510(k) would not typically identify or describe individual components of a circuit board, such as resistors, and therefore FDA would not expect changes to the resistors to be listed in the new 510(k) for a modified device because the first 510(k) would not have included information about the resistors.

If a manufacturer makes multiple changes to a device, but only one change triggers the requirement for submission of a new 510(k), the changes that do not require submission of a new 510(k) may be immediately implemented, so long as those changes can be implemented independently of changes that do require submission of a new 510(k). Any immediately implemented change should still be documented in accordance with applicable QS regulations and the manufacturer's documentation procedures. Those changes should, however, also be described in the new 510(k) for the change that does require submission.

- 10. Substantial equivalence determinations** – Manufacturers should understand that, even though they may follow this guidance and submit a new 510(k), a substantially equivalent determination is not assured. See FDA's guidance *The 510(k) Program: Evaluating Substantial Equivalence in Premarket Notifications (510(k))* (<https://www.fda.gov/downloads/medicaldevices/deviceregulationandguidance/guidancedocuments/ucm284443.pdf>) for more information on the decision-making process FDA uses to determine substantial equivalence.

V. How to Use This Guidance

This guidance uses flowcharts and text to guide manufacturers through the logic scheme we recommend to arrive at a decision on whether to submit a new 510(k) for a change to an existing device. A single logic scheme containing all the necessary steps would be large and cumbersome and could be quite daunting. Rather, for ease of use, the single scheme has been broken down into smaller sections that include:

- The main types of changes that might be made to a device (this section, Main Flowchart)
- Labeling changes (Section A, Flowchart A)
- Technology, engineering, and performance changes (Section B, Flowchart B)
- Materials changes (Section C, Flowchart C)
- Technology, engineering, performance, and materials changes for [in vitro diagnostic devices](#) (IVDs) (Section D, Flowchart D)
- Considerations for risk-based assessments of modified devices (Section E)

The main flowchart is provided in Figure 1 below and guides the manufacturer to the appropriate specific section(s) and flowchart(s) to assess their specific change(s).

When using the flowcharts, the reader should interpret “new 510(k)” as **submission of a new 510(k) is likely required** and “documentation” as **submission of a new 510(k) is likely not required, document your analysis and file it for future reference**. Please refer to Appendix C: Significant Terminology for the meaning of terms used in the guidance, including in the flowcharts.

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Note that the first question is whether the change is being made with the intent to significantly improve the safety or effectiveness of the device, for example, to significantly improve clinical outcomes, to mitigate a known risk, in response to adverse events, etc. (Figure 1 – Main Flowchart). If so, the change likely could significantly affect safety or effectiveness and submission of a new 510(k) is likely required. If not, you should continue to follow the logic scheme shown in Figure 1, below.

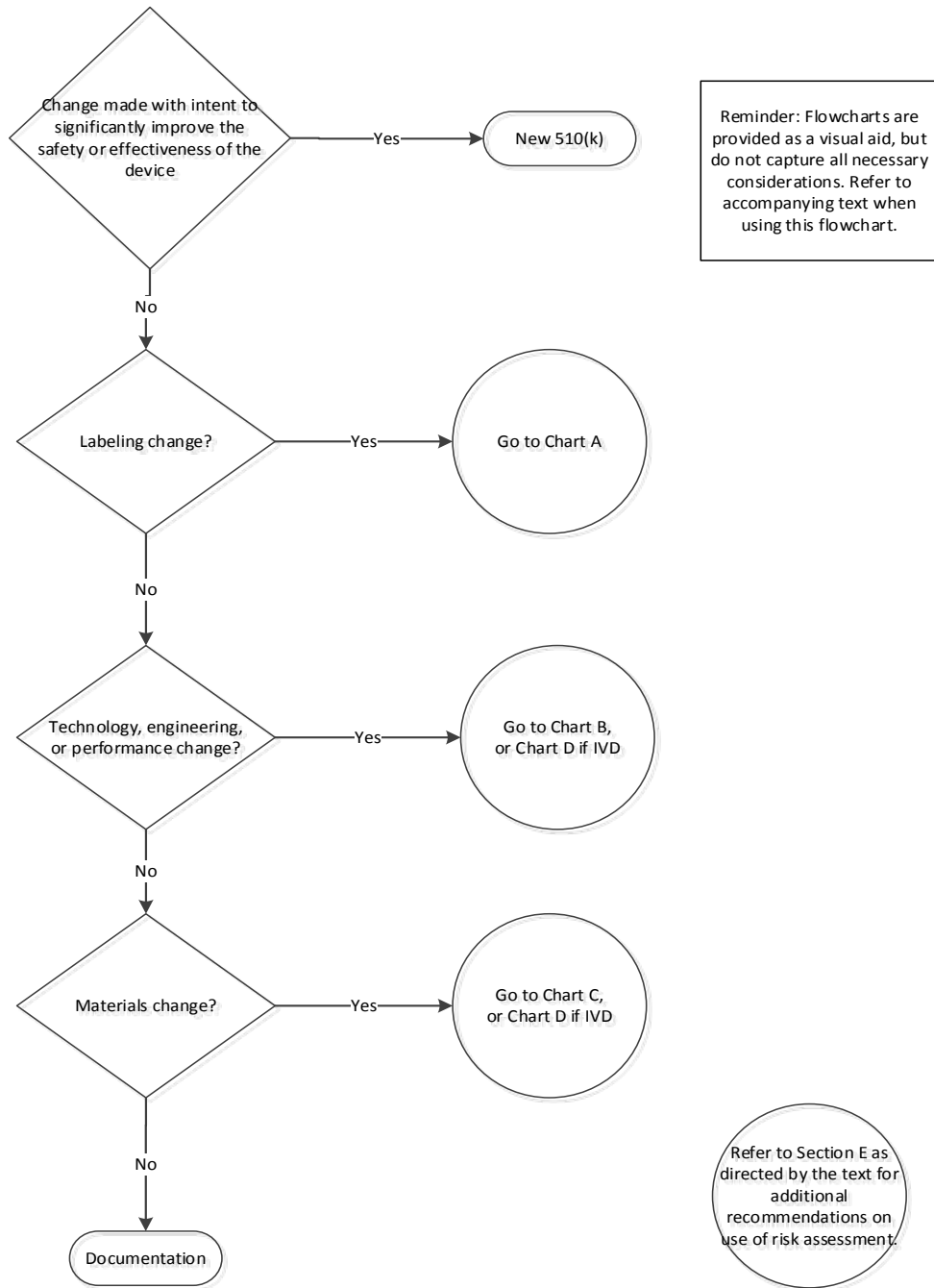


Figure 1 - Main Flowchart

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Note that sections B and C are only applicable to non-IVDs, and section D is only applicable to IVDs. All other sections apply to both IVDs and non-IVDs.

Each of the questions listed on the detailed flowcharts are identified by the flowchart letter (A through D) and a sequential number. Those questions on the main spine of the flowcharts relate to major questions to be answered. Subsidiary questions are identified by the flowchart letter, the question number, a decimal point, and another sequential number (e.g., B4.1 is a decision point containing a follow-up question that builds off a determination made in decision point B4).

Manufacturers should use the flowcharts in concert with the Guiding Principles above, the recommendations in the sections below, and the examples provided in Appendix A.

Manufacturers should follow all the applicable flowcharts and use their companion text to answer the questions posed for each individual type of change (e.g., performance change, material change) until a decision is made either to submit a new 510(k), or to document the basis for concluding that submission of a new 510(k) is not required. As mentioned above, when making the decision on whether to submit a new 510(k) for changes, the manufacturer's basis for comparison of any changed device should be the original device. Manufacturers are required to submit a new 510(k) when a change (or changes) exceeds the 21 CFR 807.81(a)(3) threshold, “could significantly affect the safety or effectiveness of the device,” or constitutes a “major change or modification in the intended use of the device.” This significant effect could be positive or negative. One must keep in mind that what may on the surface appear to be one discrete change to a device may involve multiple changes of various types.

Although this guidance does not specifically discuss manufacturing changes, a manufacturer should consider the impact of all manufacturing changes on device labeling, technology/engineering/performance, and/or materials. If the manufacturing change affects any of these three areas, manufacturers should evaluate the impact of the resulting labeling, technology/engineering/performance, or material change using the appropriate flowcharts and companion text. Specifically, consideration should be given to those devices for which manufacturing information was submitted in the most recently cleared 510(k) in order to assist in the characterization of the device and technology, such as bioresorbables, polymers, and biological fixation type devices. When manufacturing changes do not impact device labeling, technology/engineering/performance, and/or materials, there is no need to use the flow charts and their companion text to document the decision not to submit a new 510(k).

In cases with multiple changes, manufacturers should use all applicable flowcharts and companion text, including the Guiding Principles in Section IV of this guidance. Consider the following examples:

Example 1: Multiple changes caused by a manufacturing process change

A manufacturer decides to change the manufacturing process for a patient-contacting part from a machining process to a stamping process. The use of the stamping process requires a change in the grade of stainless steel and also results in a change of the dimensional tolerances. To evaluate the impact of this change, the manufacturer should use both Sections B (Technology, Engineering, and Performance) and C (Materials).

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Example 2: Multiple changes related to a change in [shelf-life](#)

A manufacturer changes one or more materials in a device to improve the shelf-life of the product. The material change also affects some of the performance characteristics, resulting in the need to update the labeling. To evaluate the impact of the change, the manufacturer should use Sections A (Labeling), B (Technology, Engineering, and Performance) and C (Materials) or D (Technology, Engineering, Performance, and Materials Changes for IVD Devices).

Changes not addressed in Sections A through D should be evaluated with a risk-based assessment using the recommendations provided in Section E. In instances where the specific flowcharts do not address a given change, Section E provides recommendations for how manufacturers should utilize risk management principles to evaluate their own specific changes and modifications. Because 21 CFR 807.81(a)(3)(i) requires submission of a new 510(k) when a change “could significantly affect safety or effectiveness,” both safety and effectiveness should be considered in evaluating a device’s risk profile, as explained in the Guiding Principles and Section E. For those circumstances where the proposed change is not addressed in this guidance or in a device-specific guidance document, manufacturers are encouraged to contact [CDRH staff](#) or [CBER staff](#).

A. Labeling Changes

As noted above, the guidance focuses on the following types of changes: labeling changes, technology, engineering, or performance changes, and materials changes. This guidance identifies several types of labeling changes or modifications to an existing device, including certain changes to the indications for use, that can have a major impact on intended use and thus require submission of a new 510(k) under 21 CFR 807.81(a)(3).² All labeling changes should be evaluated using a separate logic scheme that concentrates on changes in indications for use and applies a risk-based assessment framework for determining whether submission of a new 510(k) is required. Focusing on indications for use and using a risk-based assessment for labeling changes will also help identify those changes that are more frequently recommended for documentation only.

Flowchart A describes the logic scheme to be used when determining when submission of a new 510(k) is required for a labeling change. Changes in device labeling often pose the most difficult questions to be addressed by device manufacturers when deciding whether submission of a new 510(k) is required. Frequently, an apparently subtle change in a device’s labeling can have a significant impact on the safe and effective use of the device.

Confusion often results when discussing the distinction between “[indications for use](#)” and the “[intended use](#)” of the device. For purposes of substantial equivalence, and for the purposes of this guidance, the term intended use means the general purpose of the device or its function, and encompasses the indications for use.³ The indications for use generally describe the disease or

² Labeling changes are not the only type of changes that could result in a major change in intended use. See 21 CFR 801.4.

³ When submitting a 510(k) premarket notification to FDA for review, an applicant must submit, among other things, information concerning a device’s intended use(s), as described in the proposed labeling (21 CFR 807.92(a)).

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condition the device will diagnose, treat, prevent, cure or mitigate, including a description of the patient population for which the device is intended.⁴ The indications include all the labeled patient uses of the device. As it relates to medical devices, the indications for use statement is a factor in determining a device's intended use; however, a change in indications for use that requires the submission of a new 510(k) does not necessarily mean that the device has a new intended use (such that the device would not be substantially equivalent under section 513(i) of the FD&C Act).⁵

FDA looks to this aspect of the submission to make a substantial equivalence determination under section 513(i) of the Federal Food, Drug, and Cosmetic Act (FD&C Act), which requires FDA to evaluate whether, based on the proposed labeling, the device and a predicate device have the same intended use. If a particular labeling change results in an intended use of the device that is not the same as the intended use of the original device, the device would not be substantially equivalent. See also FDA's guidance *The 510(k) Program: Evaluating Substantial Equivalence in Premarket Notifications (510(k))* (<https://www.fda.gov/downloads/medicaldevices/deviceregulationandguidance/guidancedocuments/ucm284443.pdf>) Although, in evaluating substantial equivalence in reviewing a 510(k), FDA must determine the intended use of a device based on the proposed labeling, see 21 USC 513(i)(1)(E), FDA may consider other evidence of intended use in determining whether there has been a major change or modification in the intended use of the device under 21 CFR 807.81(a)(3).

⁴ See FDA's guidance *The 510(k) Program: Evaluating Substantial Equivalence in Premarket Notifications (510(k))* (<https://www.fda.gov/downloads/medicaldevices/deviceregulationandguidance/guidancedocuments/ucm284443.pdf>)

⁵ *Ibid.*

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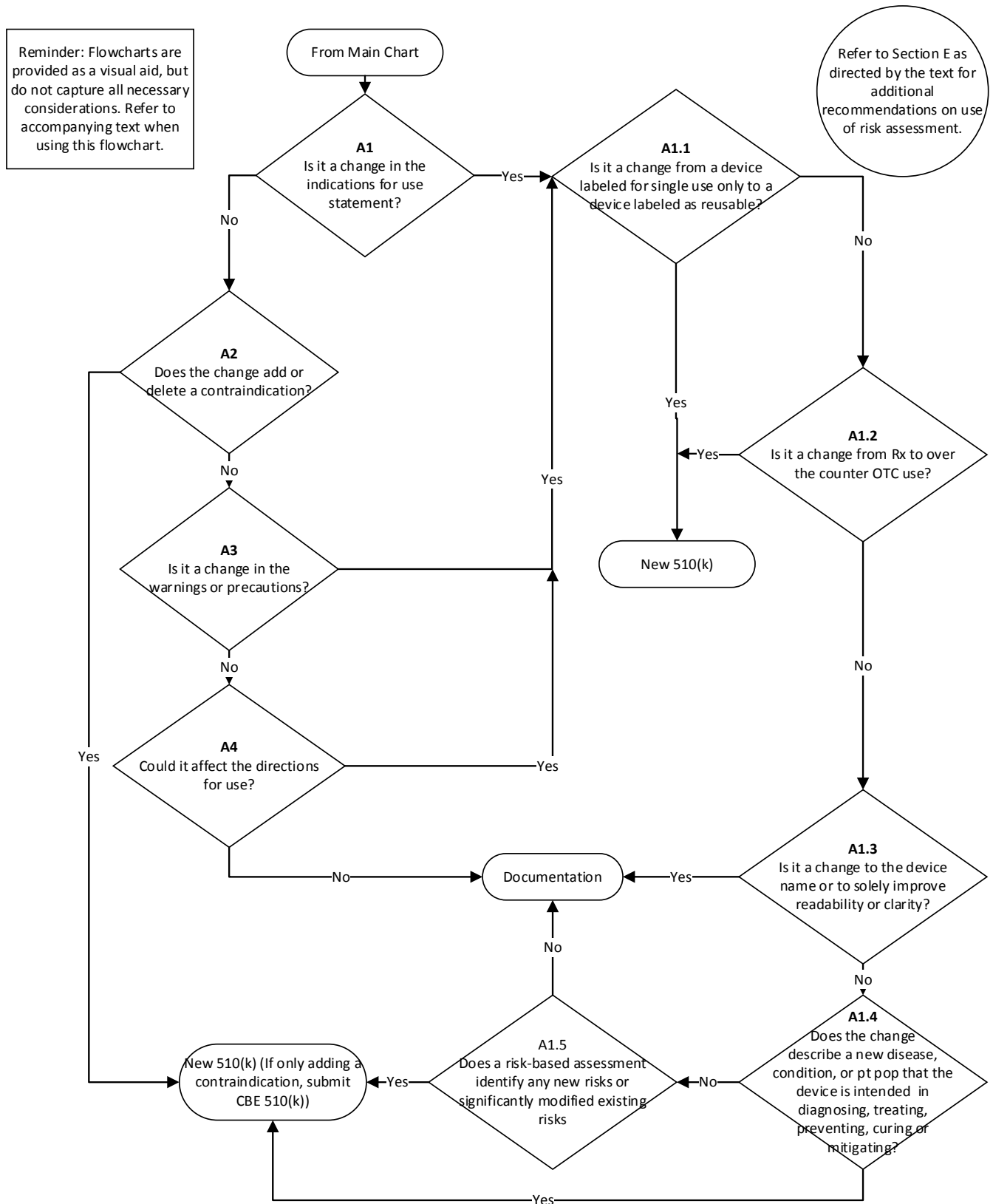


Figure 2 - Flowchart A: Labeling Changes

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- A1. Is it a change in the indications for use statement?** Changes in the indications for use statement raise more Agency concern than any other aspect of labeling. In fact, most labeling changes that affect the substance, meaning, or scope of the indications for use could significantly affect safety or effectiveness and will require submission of a new 510(k). Changes that clarify the indications without affecting the substance or meaning of the indications usually do not require submission of a new 510(k). In addition, some changes in the indications for use that limit use within the currently cleared indication may occur without submission of a new 510(k). For example, if a device was cleared for use with three specific indications and the firm decides to market the device for only two of those indications, this change would not likely require submission of a new 510(k).

If the labeling change is to the indications for use statement, proceed to **A.1.1**.
Otherwise, proceed to **A.2**.

It should be noted the decision points in **A1.1-A1.5** may apply not only to changes to the indications for use statement of the labeling, but also to changes to other sections of the labeling, such as the directions for use of the device. You should review these decision points when directed by the text of this guidance and Flowchart A: Labeling Changes.

A1.1 Is it a change from a device labeled for single use only to a device labeled as reusable?

FDA has found that the performance and risks associated with a reusable device can be significantly different from the performance and risks associated with that same device when it is labeled for single use only. Therefore, changing a device labeled for single use only to a device that is labeled as reusable typically could significantly affect the safety or effectiveness and would likely require submission of a new 510(k). Changing a device labeled for reuse to single use only, however, would likely not require submission of a new 510(k) because a single use is a limitation of the previously cleared indications for multiple uses, and the risks of single use were inherently considered within the risks of multiple uses.

If it is not this type of labeling change, proceed to **A.1.2**.

A.1.2 Is it a change from prescription (Rx) to over the counter (OTC) use?

FDA has found that the directions for use necessary for health care professionals to use a device safely and effectively can be significantly different from the directions for use necessary for lay users to use that same device safely and effectively. Therefore, changing a device labeled for prescription use only to a device that is labeled for OTC use typically could significantly affect the safety or effectiveness and would likely require submission of a new 510(k). Changing a device labeled for OTC use to prescription use, however, would likely not require submission of a new 510(k) because it is unlikely that the associated labeling changes could significantly affect the safety or effectiveness of the device.

If it is not this type of labeling change, proceed to **A.1.3**.

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A.1.3. Is it a change to the device name or a change solely to improve readability or clarity?

Changes to the device name or description that are consistent with the cleared indications for use typically do not significantly affect the safety or effectiveness and would likely not require submission of a new 510(k). Changes that are solely to improve readability or clarity that are consistent with the cleared indications for use typically do not significantly affect the safety or effectiveness and likely would not require submission of a new 510(k).

If it is not this type of labeling change, proceed to **A.1.4.**

A.1.4 Does the change describe a new disease, condition, or patient population that the device is intended for use in diagnosing, treating, preventing, curing or mitigating?

Differences in indications for use should be analyzed to explain how they are or are not critical to the intended therapeutic, diagnostic, prosthetic, or surgical use of the device, and how the differences do or do not affect the safety and effectiveness of the device. Specific changes that could significantly affect the safety and effectiveness include describing a new disease, condition, or patient population that the device is intended for use in diagnosing, treating, preventing, curing or mitigating (or an anatomical site from which a new disease, condition, or population may be inferred). The criticality of these types of changes and their direct effect on safety and effectiveness means that a change to add a new disease, condition, or patient population likely requires submission of a new 510(k).

As introduced prior to Section A.1.1., not all changes that describe a new disease, condition, or patient population that the device will diagnose, treat, prevent, cure or mitigate are necessarily made in the indications for use section of the labeling. These types of changes could also result from a change to other sections of the labeling, such as the directions for use of the device. For example, a device's directions for use may be revised from providing that the device is to be used when symptoms occur (i.e., use in the treatment of a disease or condition) to providing that the device is to be used once per day, even in the absence of symptoms (i.e., use in the prevention of a disease or condition). In this example, the patient population has changed from patients who have been previously diagnosed with a disease or condition to asymptomatic or healthy individuals. Because this change describes a new patient population, submission of a new 510(k) is likely required.

To evaluate whether a change in patient population is a new and distinguishable patient population, manufacturers should compare the demographics, diagnosis, prognosis, comorbidity, and potential for complications of the patient population described in the previously cleared 510(k) to those of the modified patient population. If the change describes a patient population with similar demographics, diagnosis, prognosis, comorbidity and potential for complications, then submission of a new 510(k) is likely not required. However, if any of these factors differ between patient populations, submission of a new 510(k) would likely be required.

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One method for determining whether changes to the demographics, diagnosis, prognosis, comorbidity, and potential for complications of the previously cleared patient populations result in a new and distinguishable patient population is to assess if the changes could significantly affect a device's risk profile through a risk-based assessment as described in A.1.5 and Section E. For example, if the most recently cleared patient population included only individuals with Stage IV carcinoma, and the modified patient population added individuals with Stage III carcinoma, a risk-based assessment considering the factors outlined in section A.1.5 could help determine whether there are any new risks or significantly modified existing risks that would require submission of a new 510(k).

If it is not this type of labeling change, you should proceed to **A.1.5**.

A.1.5. Does a risk-based assessment of the changed device identify any new risks or significantly modified existing risks?

For changes that are not addressed by the previous questions in this section, thus necessitating the use of a risk-based assessment as described in Section E, the factors discussed below should be considered as part of such an assessment for a labeling change.

As discussed in Question 1 of the Main Flowchart, if a change is intended to significantly affect safety or effectiveness, particularly those meant to significantly improve clinical outcomes, to mitigate a known risk, or in response to adverse events, that change likely requires submission of a new 510(k); this includes changes to the indications for use. For labeling changes that are not intended to significantly affect safety or effectiveness, manufacturers should consult Flowchart A and Section E and consider whether the change creates new risks or significantly modifies existing risks.

Changes to the labeling can affect a device's risk profile by affecting how, when, where, or by whom the device is used. As part of the risk-based assessment of a labeling change, manufacturers should consider whether the change could introduce human factors or usability issues that could significantly affect users' understanding of the labeling and use of the device. Changes that significantly affect a device's risk profile likely require submission of a new 510(k).

As further described in Section E, the risk-based assessment should include an analysis of both safety and effectiveness. A risk-based assessment will help manufacturers determine whether changes such as the following could significantly affect safety or effectiveness and would require submission of a new 510(k).

Changes to the type of joint, organ, bone, vasculature, or tissue applied to or interacted with, regardless of the section of labeling in which this information is contained: Although some changes to the type of joint, organ, bone, vasculature, or tissue applied to or interacted with would involve a new disease, condition, or patient population, and thus lead to a decision to submit a new 510(k) under A.1.4, a risk-based assessment would be appropriate for other changes in this category. How a change to the type of joint, organ, bone, vasculature, or tissue applied to or interacted with affects a device's risk profile depends on the specific change. For example, a change from use of a

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bone fixation system – plates, screws, and wires – in an extremity to use in the skull is likely to significantly affect the device’s risk profile and require submission of a new 510(k). Alternatively, a bone fixation system used on one type of long bone changed to use on a different type of long bone may not significantly affect the device’s risk profile and is less likely to require submission of a new 510(k).

Changes in user or use environment: How a change of this type affects a device’s risk profile depends on the differences in use environment and environmental specifications. For example, a change from use in a surgical suite to use in a hospital recovery room, both of which will have professional healthcare supervision, may not significantly affect the device’s risk profile. Similarly, changes between users with similar training on a specific device, such as changes between a general physician and a specialist using basic medical equipment may not significantly affect a device’s risk profile. However, changes from professional use to home use⁶ or hospital use to ambulatory transport are more likely to affect the device’s risk profile and require submission of a new 510(k) because the different environments have different levels of professional healthcare supervision and offer different environmental challenges, such as presence of other electronic devices that can cause electromagnetic interference, different levels of cleanliness, or shocks and vibrations associated with patient travel or ambulatory use. Similarly, changes from professional use to home use, hospital use to ambulatory transport, or between any other healthcare providers with different levels of training on specific devices are more likely to affect the device’s risk profile and require submission of a new 510(k) because the different level of training could significantly affect the safe and effective use of the device.

Changes in frequency or duration of use: Changes in the frequency or duration of use of a device include changes indicating that a device can or should be used more or less often, changes indicating that a device can perform a task or treat a condition in or for a different duration of time, or changes between periodic and continuous monitoring. Manufacturers should evaluate the effect such changes could have on the performance of a device, and whether such changes significantly affect the device’s risk profile.

Changes concerning the compatibility or interoperability of a device with other devices, components, or accessories: Two examples of such changes would include 1) changes indicating an IVD reagent for use with a new system, and 2) changes that describe how to use an infusion pump with inputs from other devices not described in the previously cleared 510(k), such as a pulse oximeter or blood pressure monitor.

To evaluate whether these changes significantly affect the device’s risk profile, manufacturers should carefully consider the following factors:

⁶ A home use medical device is a medical device intended for users in any environment outside of a professional healthcare facility. This includes devices intended for use in both professional healthcare facilities and homes. See FDA’s Home Use Devices website for more information:
<http://www.fda.gov/MedicalDevices/ProductsandMedicalProcedures/HomeHealthandConsumer/HomeUseDevices/default.htm>.

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- Differences between the other devices, components, or accessories referred to in the previously cleared indications and the ones referred to in the modified indications. Manufacturers should be able to clearly identify and analyze the risks associated with such differences, including whether the change may affect biocompatibility, performance, connectivity, etc. If the change is to indicate compatibility with a type of device, component, or accessory that was not indicated as compatible previously, that change will likely require submission of a new 510(k).
- The criticality of the other device, component, or accessory; the more critical the other device, component, or accessory is to overall system function, the more likely a labeling change regarding compatibility or interoperability could significantly affect safety or effectiveness.
- The labeling of the other device, component, or accessory. If the change is to indicate compatibility or interoperability with another device that is labeled for use with the subject device or device type, it is less likely that the change introduces a compatibility or interoperability issue that could significantly affect safety or effectiveness.

IVD manufacturers should see also FDA's guidance *Replacement Reagent and Instrument Family Policy*

(<http://www.fda.gov/downloads/medicaldevices/deviceregulationandguidance/guidancedocuments/ucm071465.pdf>).

Changes from a general use to a more specific use: These types of changes include those changes made to identify a specific use when the cleared device has a general indication for use. These changes are among the most difficult to assess. FDA's *Guidance for Industry: General/Specific Intended Use* (<https://www.fda.gov/RegulatoryInformation/Guidances/ucm073944.htm>) provides information on when a specific indication for use is reasonably included within a general indication for use for purposes of determining substantial equivalence, i.e., whether a 510(k) can be cleared or whether, instead, a PMA or De Novo request is required. The factors discussed therein – particularly those discussing the risk and public health impact of an indications for use change – may be helpful to consider in deciding whether to submit a new 510(k) for a change to an existing device, but that guidance should not be used in and of itself to justify that submission of a new 510(k) is not required. The General/Specific guidance is not intended to provide guidance on when submission of a new 510(k) is required for changes to an existing device.

If a risk-based assessment indicates that the change leads to a significant change in the device's risk profile, submission of a new 510(k) is likely required.

- A2. Does the change add or delete a contraindication?** Changes in the labeled [contraindications](#) for device use generally could significantly affect safety or effectiveness of a device and should typically be reviewed by the Agency; however, FDA recognizes that, in general, the addition of a contraindication based on new information is important to public health. Thus, FDA does not intend to object if manufacturers add new contraindications to their labeling and notify existing users of their device as expeditiously as possible whenever a pressing public health need arises. In this situation,

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the new labeling should be submitted to FDA as part of a new 510(k) that is prominently labeled “change being effected” (CBE, in Figure 2- Flowchart A). Manufacturers should ensure they are thoroughly familiar with the definition of a contraindication in such situations.

Deletion or modification of a contraindication also usually requires submission of a new 510(k) prior to effecting the change, because this type of labeling change typically changes the indications for use in a way that could significantly affect safety or effectiveness. Deletions of contraindications would expand the indications for use. For example, if a physical restraint was contraindicated for use with individuals weighing less than 100 pounds because of established life-threatening and other serious adverse events, and the manufacturer subsequently wishes to remove this contraindication, submission of a new 510(k) is likely required.

Similar to changes in indications for use, minor changes that clarify or reword a contraindication without changing the meaning of the contraindications would not typically require submission of a new 510(k).

If the change adds or deletes a contraindication, submission of a new 510(k) is likely required. Otherwise, proceed to **A3**.

A3. Is it a change in warnings or precautions? In order to facilitate a continuous upgrading in device labeling, manufacturers should monitor device usage and promptly revise the warnings and [precautions](#) section(s) based on user experience. Events that precipitate changes of this type may be those reported under the medical device reporting regulation (MDR), 21 CFR Part 803. Submission of new 510(k)s for such labeling changes are generally not required. However, to determine whether the change in warnings or precautions requires submission of a new 510(k), manufacturers should proceed to **A.1.1** and follow Flowchart A through **A.1.5**. If it is not a change in warnings or precautions, manufacturers should proceed to **A4**.

A4. Could the change affect the directions for use of the device? Device labeling may be changed for a multitude of reasons. Many labeling changes result from attempts to clarify labeling. Manufacturers should consider whether the change is intended to or could affect how the device is used in practice.

Manufacturers should evaluate labeling changes to determine whether the change affects the [directions for use](#) of the device, including IVD labeling required under 21 CFR 809.10. If the change affects the directions for use, the change should subsequently be analyzed under **A.1.1** through **A.1.5**. If the change could not affect the directions for use of the device, submission of a new 510(k) is likely not required based on the labeling change.

Examples of changes that affect the directions for use of the device, and that should be analyzed under **A1.1** through **A1.5** include:

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- Adding additional or new instructions on how to interpret diagnostic data from a diagnostic device.
- Adding a new procedural technique not described in the original labeling.
- Use of a product for a duration/frequency that is different than what is described in the labeling of the cleared device.
- Changing from labeling a device as non-sterile to labeling it as sterile or vice versa.
- Adding instructions for device use in a new patient population not described in the original indications for use.
- Adding instructions for device use in a different type of joint, organ, bone, vasculature, or tissue.

FDA believes that, if manufacturers follow this approach to changes in device labeling, only necessary new 510(k)s (those changes that could significantly impact safety and effectiveness) will be submitted, while the submission of unnecessary new 510(k)s (those that could not significantly affect safety and effectiveness) will be minimized. At the same time, manufacturers should be able to retain the flexibility to improve their labeling to assure safe and effective use of their devices.

B. Technology, Engineering, and Performance Changes

These types of changes encompass a broad span of design activities, from minor engineering changes in a circuit board layout to a change from electromechanical to microprocessor control of device function. Flowchart B illustrates the decision-making logic scheme for such technology, engineering, and performance changes to a device. These changes should be evaluated using this scheme, and then the changes should be verified and/or validated according to the QS requirements (21 CFR 820.30(i)). If the results of the verification and/or validation raise any unexpected issues, the decision of whether submission of a new 510(k) is required should be re-evaluated per B5.4.